Linear-contact restitution

PFC3D 7.0 official example | One Md=0, beta_n=0.2 subcase | 29 October 2026

PASS	All four runs compared with the published absolute-coefficient target; this is not a full 32-ball reproduction.

First-rebound height comparison

Run	Apex-derived e	Absolute error	Apex z (m)	Target
CPU dt	0.5240174395	0.0026032	0.292729420	PASS
GPU dt	0.5240174395	0.0026032	0.292729420	PASS
CPU dt/2	0.5237242200	0.0028964	0.292429882	PASS
GPU dt/2	0.5237242200	0.0028964	0.292429882	PASS

e0 = exp[-beta_n pi/sqrt(1-beta_n^2)] = 0.526620599330. e_height = sqrt[(z_apex-R)/(z0-R)], using the first rebound only. Published target: abs(e_height-e0) < 0.01. The official text says below 1%, while FISH calculates maximum absolute coefficient difference. This is not a relative 1% tolerance or automated assertion.

Official inputs and native mapping

Quantity	Retained official value / exact active mapping
Ball 8 / initial state	R=0.025 m; rho=2650 kg/m^3; center=(0.75,0.05,1) m. Zero initial velocity and spin. Free-fall clearance is 0.975 m.
Wall / contact	Fixed plane z=0; linear kn=50000 N/m; beta_n=0.2; full normal dashpot Md=0. Tangential stiffness, friction/damping and all resistance channels are zero.
Automatic damping	e_input=0.526620599330; cn=2 beta_n sqrt(m kn)=37.24965443 N s/m. Full sphere effective mass m=0.1734420944 kg. Manual overrides off; stored manual dashpots zero.
Gravity / integration	g=-10 m/s^2 throughout, including contact; local/global damping zero. Official dt=10 us; half-step=5 us. Both run to 1 s.
Complete outputs	100000 / 200000 numerical steps; every-step CSV. Native frames every 1 ms, 1001/run, 4004 total. Four loadable projects and full results retained.

One official subcase; full dashpot without clipping

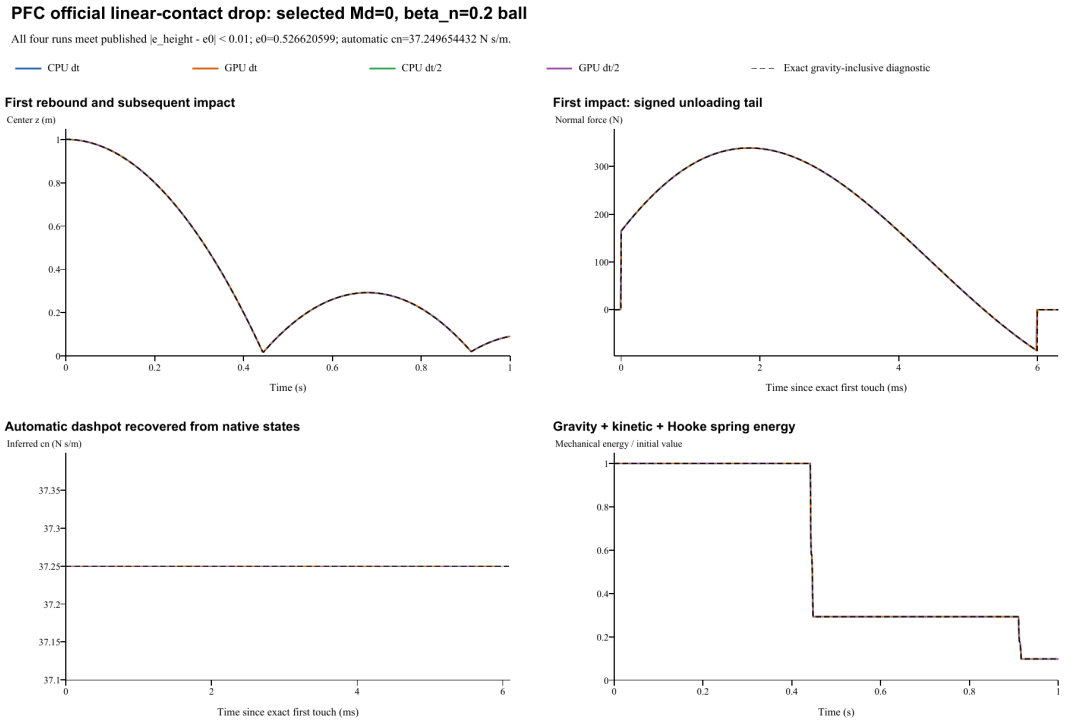
The source example contains independent parameter-sweep balls. This report uses original ball 8 only and excludes all other beta values and no-tension Md=1. Md=0 permits signed normal unloading while contact exists; its negative force tail is viscous unloading, not an adhesion model. No physical coefficient was fitted or solver source changed.

Gravity and reference provenance

The analytic e0 relation neglects gravity during contact. FunDEM keeps gravity active; an independent gravity-inclusive solution gives e_height=0.5240328046. A finite contact-duration offset can survive refinement; errors to both references increase at dt/2. Only the first rebound is measured, using every-step CSV rather than 1 ms viewing frames.

Official PFC7 example and embedded input | Source/contact audit.
Versioned documentation web-connector text snapshot SHA-256: 3dc3744ccb9cc04fb3ec1db4a2d44f27f83b50b8e187020b03c54ab80f00e7e5. PFC was not rerun locally; no Git commit/blob is invented.

Native histories and retained evidence



Native CSV: every integration step. Binary result: every 1 ms, 1001 frames/run. Two impacts occur in the 1 s window.

Half-step apex error increases: no monotonic convergence claim. Full 32-ball sweep, Md=1 clipping and PFC solver rerun are excluded.

Automatic normal damping and wall impulse

Run	Inferred cn	Min. Fn (N)	Force residual	Impulse residual
CPU dt	37.249654432	-85.27055	0	6e-13
GPU dt	37.249654432	-85.27055	5.7e-14	6e-13
CPU dt/2	37.249654432	-86.10769	0	2.6e-12
GPU dt/2	37.249654432	-86.10769	5.7e-14	2.6e-12

Force timing includes gravity: vn_assembly = vz_saved - (dt/2)(Fn/m-10); cn_inferred=(kn delta-Fn)/vn_assembly. The audit excludes near-zero denominators and checks direct force reconstruction. Coefficients are inferred from native state, not a direct output channel.

CPU/GPU whole-history maximum position / velocity / signed-force differences are 3.29e-15 m / 1.34e-12 m/s / 1.69e-10 N.

Halving dt changes apex-derived restitution by -0.0002932196. The absolute differences from the published analytic coefficient are 0.00260316 / 0.002896379. The half-step error increases; this comparison does not demonstrate monotonic convergence. Both time steps use the same physical parameters and 1 s endpoint.

The fixed wall supplies external impulse, while gravity acts throughout. The maximum residual in m(v-v0) = integral(F_contact + mg)dt is 2.62e-12 kg m/s. Sphere momentum alone is not conserved.

Mechanical energy includes gravity and spring energy once; the dashpot dissipates energy. Energy conservation and particle momentum conservation are not acceptance conditions here. Tangential force, torque and spin remain inactive.

Complete results and scope

All 4004 configured native frames, run/project manifests, four input projects, every-step particle/contact CSV and CPU/CUDA logs are retained. Real SimulationSession executes linked FunDEM 0.4.3 native libraries. Installed GUI executable byte identity is not established.

Run manifest | Analysis JSON | Independent audit | Complete native ZIP | Frame/hash audit

Verified scope: one official Md=0 linear-normal subcase, automatic damping, first rebound and gravity-aware force/impulse integration. The full 32-ball sweep, Md=1 clipping, active friction/rotation and general PFC model equivalence are outside scope. PFC is listed separately after the numeric LAMMPS demos.